

Yiwei He

Phone: 1-563-563-2910 | Email: hehegg321@gmail.com | West Lafayette, IN

EDUCATION

M.E. Industrial and Systems Engineering (GPA: 3.7/4.0)
Virginia Polytechnic Institute and State University (Virginia Tech)

Blacksburg, VA
Aug. 2021 – Dec. 2022

B.E. Industrial and Systems Engineering (GPA: 3.2/4.0)
Iowa State University

Ames, IA
Aug. 2014 – Dec. 2018

SKILLS

Research Methodologies: Controlled Experimental Design, Hypothesis Testing, Formative Study, Semi-structured Interviews, Qualitative Data Analysis, Statistical Analysis, Literature Review

Human Factors & Evaluation: Eye-Tracking, NASA-TLX, Usability Testing, Physiological Data Analysis

Technical Skills: Python, HTML, CSS, JavaScript, SPSS, R, Figma, Adobe XD, Unity, OpenAI Codex

RESEARCH EXPERIENCE

VR ATC Training Research

Virginia Tech

Blacksburg, VA
May. 2026 – Present

- Developed a WebXR prototype to investigate how immersive 3D airspace visualization can support early ATC (Air Traffic Control) training beyond static chart memorization, with emphasis on spatial understanding, altitude reasoning, and learner engagement.
- Designed an asymmetric VR/desktop training system to investigate instructor-trainee collaboration in immersive learning, implementing synchronized highlighting, shared layer states, in-VR task instructions, and scenario-based event overlays.
- Led the drafting of a conference paper that examines the prototype's design and its implications for immersive visualization, instructor-trainee collaboration, and ATC/ATM training. (Under Review)

BlendSketch

Georgia Tech

Atlanta, GA
May. 2026 – Present

- Developed an initial research prototype for BlendSketch, an asymmetric cross-reality collaboration system exploring how physical sketching and VR interaction can support shared problem-solving across different user interfaces.
- Designed and implemented prototype interactions for the interior-design scenario, including sketch-based object recognition, shared 2D/3D workspace state, and an egocentric VR review workflow for investigating mixed physical-digital collaboration.
- Conducted background research on augmented sketching, mixed-reality whiteboards, and cross-reality collaboration to inform the project's research framing, interaction design rationale, and future evaluation directions.

VR Assistant: Context-Aware IoT Notification System for Immersive Environments

Virginia Tech

Blacksburg, VA
Aug. 2022 – Nov. 2022

- The research focused on the "immersion gap" in Virtual Reality (VR), the disconnect that occurs when users are interrupted by real-world events. The goal was to design a system that allows users to monitor and interact with the real world (via IoT and messages) without breaking their sense of immersion.
- Conducted Literature review to investigate methods to enhance situational awareness in VR without compromising immersion, leveraging iterative prototyping and user testing.
- Designed and executed a controlled experiment to evaluate system efficacy by analyzing both quantitative and qualitative data.

Eye-Tracking Analysis of Selective Attention and Cognitive Workload

Virginia Tech

Blacksburg, VA
Aug. 2022 – Nov. 2022

- Designed a comparative eye-tracking study using "Spot the Difference" tasks with varying complexity levels (Easy vs. Difficult) defined by object density and spot size.
- Correlated physiological metrics (pupil size, fixation, gaze entropy) with subjective NASA-TLX survey data to validate objective markers for mental stress and temporal demand.
- Generated gaze distribution plots, fixation-over-time visualizations, and heatmaps to communicate spatial attention trends between participant groups

Expert Knowledge Elicitation for ML-Driven Equity Models

Principal Global Investors

Ames, IA
Feb. 2018 – Aug. 2018

- Designed and implemented a structured case-based interview framework to gather domain expertise, enhancing dataset quality for predictive modeling and algorithmic trading strategies.
- Collected and analyzed quantitative and qualitative data through in-depth research and interviews with equity analysts to identify key features for ML-driven equity investing models.

WORK EXPERIENCE

UX Researcher

Plano, TX

Joblogic-X

Aug. 2024 – Mar. 2025

- Executed multi-method research, including competitive analysis and user interviews, to deconstruct the U.S. affiliate marketing landscape and define data-driven localization strategies for a global platform. Led user interviews and analysis, aligning teams on key insights to drive product direction.
- Synthesized complex qualitative data through thematic analysis to establish core user requirements aligning cross-functional teams and securing executive approval for the strategic product roadmap.

Product Engineer

Des Moines, IA

Principal Financial Group

Mar. 2019 – Jun. 2021

- Led the end-to-end redesign of a global financial performance reporting system to replace an error-prone, manual process. By conducting user research and implementing automated data pipelines, the project streamlined complex data visualization for senior management and significantly reduced the operational burden on global performance teams.
- Conducted user research through semi-structured interviews with international performance teams (North America, South America, and SE Asia) to map existing workflows, identify pain points in data manipulation, and document artifact usage.
- Designed and prototyped an interactive dashboard utilizing Power BI, incorporating advanced data filtering (slicers) and historical trend visualizations to facilitate high-level decision-making for senior management.